

Blockchain HW2

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1. Create your own token and [transfer 100 tokens to TA](#)

Two side-by-side screenshots of a token contract interface. The left screenshot shows the initial state with a balance of 0 ETH and a total supply of 500,000,000. The right screenshot shows the state after a transfer, with a balance of 100 tokens for the specified address.

2. Write a function similar to the faucet, and you can add some variables for it.

Two side-by-side screenshots of a token contract interface. The left screenshot shows the initial state with a balance of 500,000,000. The right screenshot shows the state after a faucet transaction, with a balance of 499,999,500.

程式解釋：

function faucet(address from, address to) public returns (bool success)

接受兩個地址參數 from 和 to，並且傳回一個布林值表示操作是否成功。

```
require(from == contractCreator, "Only the contract creator can use the faucet.");
```

要求 from 必須是合約創建者

```
require(block.timestamp > lastFaucet[to] + 30 seconds, "Please wait for 30 seconds before requesting again.");
```

同一個位址在 30 秒內只能請求一次代幣。lastFaucet[to]記錄了該位址上次請求的時間戳，block.timestamp > lastFaucet[to] + 30 seconds 確保距離上次請求已經過去了至少 30 秒才能讓該位址再次請求代幣。

```
balances[from] = safeSub(balances[from], 500);  
balances[to] = safeAdd(balances[to], 500);
```

```
emit Transfer(from, to, 500);
```

從 from 轉移 500 代幣到 to

```
lastFaucet[to] = block.timestamp;
```

轉移代幣後更新 to 的時間戳